WSABI Summer Research Lab **Summer 2026**

Lab 12: Empirical Bayes

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12.1 Predicting 2021 Batting Averages

The 2020 MLB season was shortened, making player batting averages especially noisy. Use the supplied batting-average data file to estimate 2020 underlying batting ability and evaluate the estimates against 2021 performance.

Empirical Bayes is Bayesian updating with an estimated prior. Here, the estimated prior comes from the 2020 population of related players.

For player i, use the model

$$X_i \mid \mu_i \sim \mathcal{N}(\mu_i, \sigma_i^2), \quad \sigma_i^2 = \frac{C}{N_i},$$
$$\mu_i \sim \mathcal{N}(\mu, \tau^2).$$

This lab uses the method-of-moments estimates rather than numerically maximizing the marginal log likelihood. Estimate the hyperparameters from the 2020 population as follows:

$$\widehat{\mu} = \frac{\sum_i H_i}{\sum_i N_i},$$
$$\widehat{C} = \widehat{\mu}(1 - \widehat{\mu}),$$
$$\widehat{\tau}^2 = \max\left\{0, \mathrm{Var}(X_i) - \mathrm{Mean}\left(\frac{\widehat{C}}{N_i}\right)\right\}.$$

Then

$$\widehat{\lambda}_i = \frac{\widehat{\tau}^2}{\widehat{\tau}^2 + \widehat{C}/N_i},$$
$$\widehat{\mu}_i^{\mathrm{EB}} = \widehat{\mu} + \widehat{\lambda}_i(X_i - \widehat{\mu}).$$

12.1.1 Task 1: Build the Empirical-Bayes Estimates

Use all players with at least one 2020 at-bat when estimating the 2020 population distribution. Do not filter out low-2020-at-bat players; they are where shrinkage matters most. Calculate the MLE, empirical-Bayes estimate, sampling variance, posterior variance conditional on $(\widehat{\mu}, \widehat{\tau}^2)$, and shrinkage weight for every player. Verify that posterior precision equals data precision plus prior precision.

12.1.2 Task 2: Make Shrinkage Visible

Create an arrow plot from each player's MLE to their empirical-Bayes estimate. Use color, faceting, or labels to show how movement depends on:

- distance from the league mean;
- number of 2020 at-bats;
- the shrinkage weight $\widehat{\lambda}_i$.

12.1.3 Task 3: Evaluate Future Prediction

Use 2021 batting average as the held-out outcome, restricting this evaluation step to players with at least 100 at-bats in 2021. Compare the MLE, complete-pooling estimate, and empirical-Bayes estimate using:

1. overall RMSE;
2. RMSE for low- and high-at-bat players;
3. a plot of prediction against 2021 batting average.

Do not use 2021 data to estimate any 2020 model parameter.

12.1.4 Task 4: Exact-Rate Empirical-Bayes Bridge

Translate the estimated league mean and between-player variance into a Beta distribution with the same mean and variance:

$$s = \frac{\widehat{\mu}(1-\widehat{\mu})}{\widehat{\tau}^2} - 1,$$
$$\widehat{\alpha} = \widehat{\mu}s, \quad \widehat{\beta} = (1-\widehat{\mu})s.$$

Use this estimated prior with the exact Binomial likelihood to calculate Beta-Binomial empirical-Bayes posterior means. Compare these with the Normal–Normal estimates. Where does the approximation matter most?

12.1.5 Task 5: Sensitivity and Uncertainty

Refit the estimates after multiplying $\widehat{\tau}^2$ by 0.5 and by 2. Explain how the estimates and out-of-sample RMSE change. What does this reveal about the role of estimated between-player variation?

Ordinary empirical Bayes treats the estimated hyperparameters as fixed after fitting them. Explain why this can make its uncertainty intervals too narrow.

12.2 Pooling Field-Goal Rates

The batting-average example pools players who are treated as draws from one related population. Field goals show why that assumption matters. A short attempt and a long attempt are both field goals, but they should not necessarily share the same prior make rate.

Use `12_field-goals.csv` to check whether one common prior makes sense for NFL kickers. In this file, `ydl` is a distance proxy, with larger values corresponding to longer attempts.

12.2.1 Your Task

Create three distance groups:

- short: `ydl <= 20`;
- medium: `20 < ydl <= 35`;
- long: `ydl > 35`.

Aggregate attempts by kicker and distance group, keeping kicker-groups with at least 5 attempts. For each distance group, estimate a Beta prior from the kicker make rates, then compute the Beta-Binomial empirical-Bayes posterior mean

$$\widehat{p}_{kg}^{\text{EB}} = \frac{\text{made}_{kg} + \widehat{\alpha}_g}{\text{attempts}_{kg} + \widehat{\alpha}_g + \widehat{\beta}_g}.$$

Make a plot comparing raw and empirical-Bayes make rates by distance group. Then repeat using one common prior for all field-goal attempts. Contrast the two empirical-Bayes results: which kickers or distance groups move differently, and why? Argue which pooling strategy is more defensible and what it says about exchangeability.